

riskweightedassets — complete documentation

Snapshot timestamps — 1.2.3

The public `official_snapshot()` helper previously parsed an ISO knowledge-time string without an explicit format. Under affected R versions this truncated 2026-08-31T23:59:59 to midnight and could omit same-day records. The KSA reference case selected zero instead of 30 business-indicator rows.

The helper and `calculate_tables()` now share one internal UTC parser. Accepted inputs are POSIXt instants, UTC strings with T or space separators, optional fractional seconds and optional trailing Z, and date-only values (midnight UTC). Missing, malformed and non-scalar values raise a structured configuration error. Function signatures, dependencies and source data are unchanged.

Regression tests compare the implicit `run_config` snapshot with an explicit 23:59:59 UTC snapshot, retain all 30 expected rows, verify midnight selection, exercise the supported timestamp formats and reject invalid values. Caller-owned tables are unchanged.

Canonical default flags — 1.2.2

The technical representation of a default must never change its regulatory result. This patch replaces the inconsistent SA and IRB interpretation with one internal, fail-closed regulatory boolean contract in both Python and native R.

Accepted scalar values

Meaning	Native inputs	Text inputs (case-insensitive; whitespace trimmed)
True	boolean true, integer 1, floating-point 1.0	TRUE, 1, 1.0, YES, JA
False	boolean false, integer 0, floating-point 0.0	FALSE, 0, 0.0, NO, NEIN

Missing values (None/NULL/NA/NaN), empty strings, UNKNOWN, DEFAULT, Y, N, 2, -1 and non-finite numbers fail with `default_flag must be an explicit boolean`; they are not treated as performing.

One status for all calculation paths

After official bitemporal selection and designations, `exposure_lot.default_flag` is normalised on a copy of the selected snapshot, before either reporting view is run. SA, IRB, expected loss and supporting-factor eligibility use that same boolean. Credit calculations invoked internally without the public snapshot preparation use the same normalisation helper; already canonical flags are reused.

No caller-owned table or historical input workbook is mutated. The normalised result drives actual KSA, the IRB SA-shadow, Standardised TREA and the output floor. `SA_Detail` now includes the additive boolean `defaulted`, matching `IRB_Detail`. Other convenience booleans are outside this patch.

Regression evidence and independent references

Both languages consume byte-identical `regulatory_boolean_cases.csv` test vectors, including native boolean/integer/float, text, missing and invalid groups. Tests use explicit 2026-08-31 / 23:59:59 UTC snapshot times.

Public `calculate_tables()` tests compare mixed-representation portfolios with canonical-boolean portfolios in both applied and fully-loaded views. Every test exposure has corporate CQS 3, EAD EUR 1,000,000, no adjustments, no CRM and supporting factor 1. Other capital modules are neutralised in these reference cases so that the credit and floor expectations are independent:

- Default: SA RW 1.50; SA RWEA and shadow RWEA EUR 1,500,000 per exposure.
- Performing: SA RW .75; SA RWEA and shadow RWEA EUR 750,000 per exposure.
- Pure KSA: $RWEA_KSA = U_TREA = TREA = \text{sum of the exposure RWEA}$.
- FIRB default with LGD .40: $K/RW/RWEA = 0$; effective PD = 1; EL EUR 400,000.

- Binding floor at .725: default TREA EUR 1,087,500 per exposure. The former representation-only discrepancy of EUR 543,750 disappears.
- The performing IRB cases also bind the floor and retain effective PD .0005.
- Every invalid vector is rejected through the public API in both KSA and IRB cases.
- A Python call-count assertion verifies one raw-status normalisation per exposure shared by both reporting views, not one interpretation per credit path.

The prior FIRB, AIRB, performing, retail and complete-profile tests remain release gates. Input resources and reference parameters are unchanged; no golden values are relaxed to conceal a discrepancy.

Migration

No public function signature changes and no new dependencies. Use the same canonical exposure input column. Replace missing/ambiguous flags with a documented true/false value before calculation. Existing accepted values remain accepted. Version 1.2.1's explicit default `lgd_treatment` formula contract is unchanged. Recompute portfolios that previously used decimal-number or decimal-string default flags; historical bundled outputs remain historical.

IRB default correction — version 1.2.1

Regulatory rule and scope

CRR Article 153(1)(ii) distinguishes supervisory LGD from own estimates; Article 154 governs retail, and Article 158(5) governs expected loss. For a defaulted supervisory-LGD (FIRB) exposure, K , RW and $RWEA$ are zero, while $EL = \text{effective PD} \times LGD \times EAD = LGD \times EAD$. $ELBE$ is irrelevant. For own LGD estimates (AIRB and supported retail), $K = \max(LGD - ELBE, 0)$ and $EL = ELBE \times EAD$. Supporting-factor eligibility rules are unchanged.

Source: CRR consolidated text, Articles 153, 154 and 158. This implements the previously documented distinction, not a new regulatory assumption.

Input and API contract

The portfolio resolves LGD treatment from `irb_approach` and `irb_subclass`: $FIRB \rightarrow SUPERVISORY$; $AIRB \rightarrow OWN_ESTIMATES$. Non-retail subclasses supported by this formula path are `CORPORATE`, `INSTITUTION` and `SOVEREIGN`. `RETAIL_RESIDENTIAL`, `RETAIL_QRRE` and `RETAIL_OTHER` require `AIRB / OWN_ESTIMATES`. This mapping is a calculation contract, not a certification of regulatory permission to use AIRB. Institution-specific approach permissions, LGD input values and existing floors remain the user's responsibility.

Unknown approaches/subclasses and FIRB-retail combinations fail closed. `SLOTTING` is not implemented as a separate calculation path in this library: it is now explicitly rejected instead of silently using the corporate formula. No slotting factors, correlations or maturity rules are introduced by this fix. Performing maturity adjustments and the retail no-maturity-adjustment rule remain unchanged.

`default_flag` must be a recognised explicit boolean. If true, effective PD is 1 regardless of the supplied PD (which must still be in $[0,1]$). If false, $PD = 1$ (including after a PD floor) is rejected. Detail results retain `pd_input` alongside effective `pd`, defaulted and `lgd_treatment`. Caller data are not modified.

The public `irb_capital_requirement()` adds the optional `lgd_treatment` argument. For defaulted calls it is **required**, with value `SUPERVISORY` or `OWN_ESTIMATES`. Missing/unknown values raise an error rather than guessing. Existing performing calls retain their signatures/defaults and arithmetic; R appends the argument after parameters to preserve positional compatibility. The helper returns K , not total capital or a regulatory eligibility decision.

Independent acceptance cases

For LGD .40, EAD 1,000,000, default true, supporting factor 1:

Treatment	ELBE	K	RW	RWEA	EL
FIRB	.10	0	0	0	400,000
FIRB	.80	0	0	0	400,000
AIRB / retail	.10	.30	3.75	3,750,000	100,000
AIRB / retail	.40	0	0	0	400,000
AIRB / retail	.50	0	0	0	500,000

With eligible coverage 250,000, the FIRB case has shortfall 150,000 and excess 0. Coverage comes from exposure_lot, not duplicate ad-hoc IRB columns. Tests exercise the public `calculate_tables()` path in both applied and fully-loaded views, EL/RWEA aggregates, CET1 shortfall deduction and the TREA/output-floor chain. Both languages use these independently specified values, plus normal-distribution reference calculations for performing FIRB/AIRB and retail.

Reference data and migration

Synthetic input resources remain unchanged; they are not rewritten to conceal the defect. Historical bundled outputs are historical snapshots, not new-version acceptance evidence. Recalculate affected default portfolios with version 1.2.1. The intentional differences from 1.2.0 are supervisory-default RWEA and EL, their capital consequences, normalised default PD, additive audit columns and rejection of ambiguous/unsupported inputs. Existing aggregation and own-funds rules are retained; this patch does not redesign the coverage allocation model.

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1. Purpose and status

`riskweightedassets` is an independently installable, native R implementation of the RiskDataScience risk-weighted-assets reference engine. It calculates CRR-oriented Pillar 1 exposure amounts, own-funds and prudential constraints, IRRBB/CSRBB and ICAAP views from a governed canonical data model. It includes realistic synthetic bank data and does not require Python at runtime.

The package supports research, education, prototyping and independent model validation. It is not a certified reporting solution and does not replace legal assessment, permissions, institution-specific mapping, model validation or supervisory approval.

2. Installation

R 4.1 or newer is required. Install a built source package with:

```
install.packages("riskweightedassets_1.2.3.tar.gz", repos = NULL, type = "source")
library(riskweightedassets)
```

After CRAN acceptance use `install.packages("riskweightedassets")`.

Runtime dependencies are `digest`, `jsonlite`, `openxlsx`, `readxl`, `utils` and `yaml`. They are ordinary CRAN packages. Python, NumPy and downloaded regulatory material are not runtime dependencies.

3. Quick start

An in-memory reference calculation has no output side effects:

```
tables <- generate_synthetic_tables(bank_profile = "MID_SIZE_UNIVERSAL")
result <- calculate_tables(tables)
print(result)
unlist(result$metrics[c("RWEA_KSA", "RWEA_IRB", "TREA", "CET1_RATIO")])
```

The equivalent spreadsheet workflow is:

```
dataset <- generate_synthetic_dataset(
  root = file.path(tempdir(), "rwa-runs"),
  as_of = as.Date("2026-08-31"),
  version = "v1.0.0",
  bank_profile = "KSA_BANK"
)
report <- validate_dataset(dataset)
result <- calculate_dataset(dataset)
```

4. Public API

The package exports 77 documented functions. The controlled lifecycle remains:

- `calculate_tables()` calculates a named list of 68 canonical data frames and returns applied and fully-loaded results without writing files.
- `validate_dataset()` validates 16 canonical input workbooks without a run.
- `calculate_dataset()` performs the complete controlled spreadsheet run and writes six output workbooks and a JSON manifest.
- `generate_synthetic_tables()` materialises either reference profile in R.
- `generate_synthetic_dataset()` materialises a profile as canonical XLSX.
- `create_workspace()` creates a writable self-contained reference workspace.
- `default_workspace()` resolves the environment-configured default without writing it.
- `list_reference_profiles()` and `list_reference_datasets()` expose inventory.
- `regulatory_sources()` exposes official URLs and archival source hashes.

The bank-analyst layer additionally exposes 38 individual regulatory and risk formula functions, nine domain-analysis functions, applied/fully-loaded metric and control accessors, result-table accessors, the formula and rule-set catalogues, canonical schemas and bitemporal snapshots. Regulatory parameters are inspectable and may be overridden non-destructively only with a reason and approval reference; the result preserves the old/new-value audit trail. A structurally changed formula remains a versioned code and test change rather than an ungoverned runtime expression. The complete inventory is in `docs/ANALYST_API.md`.

The principal S3 result class is `rwa_calculation_result`. Its fields are `status`, `run_id`, `engine_version`, `rule_set_id`, `metrics`, `controls`, `validation`, `output_dir`, `output_files`, `results`, `parallel_results`, `parallel_metrics`, `parallel_controls` and `parameter_overrides`. `rwa_validation_report` converts to a six-column data frame. The condition hierarchy distinguishes configuration, parameter, resource, validation and calculation errors.

5. Data contract

The canonical model contains 68 tables across 16 input workbooks. Each data sheet has one Excel table named `tbl_in_<logical-name>`. Schema identifiers are stable and case-sensitive.

All historised tables begin with `record_id`, `business_key`, `version_no`, `as_of_date`, `valid_from`, `valid_to`, `known_from`, `known_to`, `is_official` and `record_status`. Business validity and knowledge validity are end-exclusive. An official snapshot retains the most recent eligible record version per business key. An approved official designation can identify a specific record version explicitly.

The domains cover run governance, entities and consolidation, parties and ratings, contracts/facilities/exposures, SA and IRB credit, collateral and guarantees, derivatives and SFT, CCP/CVA, securitisation, market risk, operational risk, own funds, IRRBB cash flows and curves, ICAAP, model/formula registry, mappings, parameters and legal-source lineage.

Dates are represented as `Date`, knowledge timestamps as UTC `POSIXct`, and monetary/rate values as `double`. Missing Excel cells become typed `NA`. Boolean decoding is explicit; the canonical fixture regression test specifically guards against spreadsheet readers treating OOXML false (0) as true.

6. Validation

Validation precedes all calculation. It covers expected files, worksheets and Excel table names; required values; record uniqueness; interval logic; selected foreign keys; numeric bounds; parameter uniqueness; run configuration; tranche bounds; and curvature revaluation completeness.

Each issue has severity, stable code, table, row reference, field and message. Errors prevent calculation. Warnings are retained in the audit result. This technical validation is necessary but does not establish the completeness or correctness of a real bank's source data.

7. Calculation architecture

Configuration and regulatory values are loaded first. The contract layer normalises and validates inputs and creates one official bitemporal snapshot. The same snapshot is calculated under the applied 2026 rule set and a parallel fully-loaded rule set. Engines add result tables and metrics to a controlled bundle. Reconciliations and formula-registry checks run before output metadata is added.

Pure formula functions are separated from portfolio orchestration. Excel I/O is a boundary layer, not a business-logic dependency. Package resources are immutable. Only destinations explicitly selected by the caller are writable.

8. Pillar 1 methodology

8.1 Standardised credit risk

SA EAD separates on-balance-sheet and off-balance-sheet amounts. Off-balance amounts use annex-class conversion factors. Exposure class, credit quality, default, short-term, retail, transactor, specialised-lending, currency and real-estate attributes select the regulatory risk weight. Real-estate logic uses property type, IPRE/ADC/completion status, value, liens and ETV. CRM applies exposure/collateral haircuts, maturity adjustment, FX haircut and funded or unfunded protection. Supporting factors are applied only after eligibility.

8.2 IRB credit risk

Corporate and retail functions apply PD/LGD floors, asset correlations, confidence quantile, maturity treatment, default handling and financial-sector multipliers. EAD times unexpected-loss capital times the Pillar 1 multiplier produces RWEA. Expected loss reconciles with eligible provisions. The engine also calculates every exposure under SA for output-floor purposes.

8.3 Counterparty, CVA and related risks

SA-CCR calculates replacement cost, PFE multiplier, supervisory add-ons and netting-set EAD. SFT uses a comprehensive exposure/collateral method. CCP trade/default-fund calculations remain distinct. BA-CVA is official where configured and SA-CVA is a parallel sensitivity-based view. Settlement-delay and large-exposure excess charges are calculated separately.

8.4 Securitisation

The approach hierarchy selects SEC-IRBA, SEC-SA or SEC-ERBA based on available inputs and eligibility. SSFA methods use pool capital, delinquency, tranche attachment/detachment and supervisory parameters. ERBA uses rating, maturity, seniority and STS. Regulatory floors and caps are enforced.

8.5 Market and operational risk

The applied rule set reports the configured legacy market-risk requirement and produces FRTB SA/IMA parallel views. FRTB SA combines SBM delta/vega/curvature, DRC and RRAO under correlation scenarios. The fully-loaded rule set explicitly selects FRTB. Operational risk derives BI components from three-year accounting measures, applies marginal coefficients and the configured EU ILM.

8.6 TREA and output floor

Unfloored TREA aggregates credit RWEA and equivalents of the included capital requirements. Shadow TREA uses full SA credit. The applicable transitional or fully-loaded factor floors the result. Floor uplift is allocated to components and reconciled. Pillar 2 and economic-management equivalents are prohibited from being double counted in legal TREA.

9. Capital and prudential constraints

Own funds combine eligible CET1, AT1 and T2 instruments/components and apply deductions, prudent valuation, NPE backstop and IRB shortfall/excess. Capital ratios and headroom distinguish minimums, Pillar 2 requirement, combined buffer and Pillar 2 guidance. Leverage, MREL and TLAC use their own denominators and eligibility rules. Management equivalents remain transparent bridge views.

10. IRRBB, CSRBB and ICAAP

IRRBB assigns contractual and behavioural cash flows to repricing bands, interpolates currency curves and applies six versioned shock shapes. It reports currency and aggregate EVE/NII impacts, worst losses and outlier ratios. CSRBB is a separate stress loss. The stochastic EVE approximation reports 99% VaR and expected shortfall from 5,000 versioned rate shocks. The canonical path is shared with Python 1.0.0 for cross-language parity and does not alter the user's global R random state.

ICAAP calculates standalone economic capital after provisions, model risk and non-diversifiable add-ons. A validated correlation matrix aggregates diversifiable capital, subject to a diversification cap. Economic capacity applies eligibility, haircut and reserve policy. Normative projections roll CET1, TREA and leverage exposure through scenarios and expose minimum headroom.

11. Controls and auditability

Reference runs pass controls for SA and TREA reconciliation, non-negative EAD, capital-stack reconciliation, correlation symmetry/PSD, behavioural weights, formula registration, official market regime, EU ILM, non-double-counting and floor allocation. Each result row carries run, version, date, rule-set, view and official-status metadata. The manifest fingerprints inputs, code and run context. The audit workbook retains validation, reconciliations and lineage.

12. Synthetic profiles

MID_SIZE_UNIVERSAL covers the complete mixed SA/IRB and complex-product engine. Its reference SA RWEA is EUR 5,401,278,717.59, IRB RWEA is EUR 901,108,184.57 and TREA is EUR 8,691,246,387.71.

KSA_BANK applies declarative transformations to create an SA-only institution while retaining capital, IRRBB and ICAAP. Its reference SA RWEA is EUR 9,752,217,943.87 and TREA is EUR 11,040,467,943.87.

These values are regression fixtures, not industry benchmarks. Both profiles contain only synthetic names, contracts and exposures.

13. Testing and parity

Tests cover formulas, invariants, parameter strictness, classes, resources, schema and boolean integrity, date shifting, profile transformation, workbook round trips and both complete calculations. All 63 metrics and 34 result tables are classified. Numerical parity uses absolute tolerance `max(1e-5, abs(expected) * 1e-12)`; discrete values must be exact.

Release checks are:

```
roxygen2::roxygenise()
devtools::test()
devtools::check()
```

```
R CMD build .
```

```
R CMD check --as-cran riskweightedassets_1.2.3.tar.gz
```

Linux, Windows and macOS CI plus Win-builder and R-hub release/oldrel/devel checks are required before CRAN submission.

14. Regulatory sources and intellectual property

`regulatory_sources()` lists official source URLs and archival checksums. No downloaded regulation, guideline, PDF, webpage or third-party standard is included in the source repository or package. Users must retrieve and evaluate current official texts independently.

Original code, synthetic data and documentation are GPL-3. External materials remain subject to their own rights. The package performs no telemetry and no automatic source download.

15. Limitations and production adoption

The package does not supply institution-specific source-system adapters, regulatory permissions, national-option decisions, model approvals, access control, operational scheduling, backups, sign-off workflow or report filing. A bank adopting it must independently govern all of those controls, reconcile input and output to authoritative systems, validate every material model and maintain legal/configuration updates.

16. Legal contacts

- Imprint: <https://riskdatascience.net/impressum/>
- Privacy policy: <https://riskdatascience.net/datenschutzerklaerung/>
- Issues: <https://github.com/rds0001/risk-weighted-assets-r/issues>

This software is supplied without warranty and does not constitute legal, supervisory, accounting, tax or investment advice.

Credit supporting factors (1.2.0)

Four new public functions cover SME, infrastructure, combined support and complete IRB RWEA.

Detailed supporting-factor reference

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Scope and calculation

The SA and IRB paths can apply Article 501 SME and Article 501a infrastructure support separately or together. IRB calculation is:

$$\text{RWEA_before} = \text{EAD} * \text{RWA_MULTIPLIER}[\text{PILLAR1}] * \text{K RWEA_after} = \text{RWEA_before} * \text{SME_factor} * \text{infrastructure_factor}$$

K, PD, LGD, EAD, expected loss, shortfall and excess are not reduced by this step. The pre-existing SME correlation adjustment is a different mechanism.

The SME helper uses the weighted Article 501 formula: the first EUR 2.5 million of E* at 0.7619 and the remainder at 0.85, divided by E*. E* must be positive and determined under the Article's institution-group/connected-client rules, including residential collateral exclusions and the fallback when the initially calculated amount is zero. Do not substitute individual EAD or annual turnover. Infrastructure support is 0.75 by default. Both factors can apply together.

Values are in the versioned SUPPORTING_FACTOR parameter rows: SME_THRESHOLD_EUR, SME_LOWER, SME_UPPER, SME_SALES_MAX_EUR, INFRASTRUCTURE. Use existing controlled parameter overrides for sensitivities; do not edit code. An older dataset needs these explicit parameter rows only if claiming new support. They are materialized automatically when generating a new dataset.

Public formula functions

- `sme_supporting_factor(total_amount_owed_eur, parameters=...)`: weighted factor.
- `infrastructure_supporting_factor(parameters=...)`: configured infrastructure rate.

- `apply_credit_supporting_factors(rwea, sme_factor=1, infrastructure_factor=1)`: applies both components once.
- `irb_risk_weighted_assets(ead, capital_requirement, ..., parameters=...)`: complete RWA amount from unadjusted K.

Python uses keyword-only optional arguments. R allows named arguments. These are numeric building blocks, NOT eligibility certificates. Amounts supplied to the SME helper are EUR. Other pure amount calculations preserve the input unit.

Input contract and eligibility

Optional additive fields on both `sa_classification` and `irb_parameter`:

Field	Meaning
<code>sme_supporting_eligible</code>	Explicit attestation that ALL applicable Article 501 conditions hold
<code>infrastructure_supporting_eligible</code>	Explicit attestation that ALL applicable Article 501a conditions hold
<code>sme_total_amount_owed_eur</code>	Reviewed E* in EUR; required when SME support is requested
<code>supporting_factor_reference</code>	Non-empty bank-owned evidence/checklist reference
<code>supporting_factor_approved_by</code>	Non-empty governance approval reference

`irb_parameter` additionally accepts `supporting_factor_type` and `supporting_factor`, already present in the SA input contract. The type can be NONE (derive from flags), SME, INFRASTRUCTURE or SME_INFRASTRUCTURE. Leave the numeric factor blank to derive it. If supplied, it must match the calculated product; it is not an unrestricted discount override.

The engine rejects defaulted exposures, invalid/missing numeric values and unsupported classes for requested relief. SME support excludes ADC and checks available turnover against the configured SME ceiling. Infrastructure support requires an eligible corporate class. Context is checked again against the selected exposure snapshot, including party and real-estate data.

A flag attests all remaining qualitative criteria, including the applicable environmental conditions for newer infrastructure lending. The engine cannot verify bank contracts or external legal evidence. Approval text is an audit reference, not an authentication/authorization system. Institutions remain responsible for independent eligibility review and their own access controls.

SA comparison and IRB inputs are independent: never silently copy a supplied SA factor to IRB. Populate both eligible paths explicitly when both qualify.

Compatibility and audit output

Old Excel workbooks remain readable: only the new columns are filled if absent. Old IRB inputs receive no reduction. Invalid explicitly supplied values are not silently replaced. Historical SA factors without new eligibility data retain their old arithmetic, with status `LEGACY_SA_UNVERIFIED` and a validation `WARNING_LEGACY_SA_SUPPORTING_FACTOR`. This preserves reference results without claiming that historical eligibility has been established. Do not use this compatibility path as a substitute for reviewing new production eligibility.

`SA_Detail` and `IRB_Detail` expose the overall factor, component factors, type, status, evidence and approver, support formula ID, pre-support RWEA and relief. Legacy SA factors are not decomposed into falsely certified component factors. IRB's existing `rw` remains pre-support; `effective_rw` includes the factor. `IRB_Detail` also exposes `sa_comparison_supporting_factor` and `supporting_factor_path_difference`. A difference is diagnostic, not an automatic error: eligibility may differ by approach and legacy SA data can be unverified.

U-TREA uses adjusted actual IRB/SA RWEA. S-TREA uses the independently adjusted SA comparison. The existing output floor is then applied once. Neither TREA nor the floor uplift is multiplied again by a support factor. A binding floor can absorb some or all of an IRB-only reduction. Applied and fully-loaded views use the same controlled exposure data with their respective rule contexts.

The input tables retain their bitemporal metadata. New disclosures are additive; no tables or metrics are removed. Historical bundled workbooks/output runs remain unchanged reference artifacts, not recomputed 1.2.0 runs. Generated new workbooks use the extended effective schema.

Verification and limits

Focused tests cover numeric boundaries and non-finite values, combined factors, missing evidence, ineligible/defaulted/ADC exposures, unchanged K/EL, old inputs, parameter sensitivity, Excel round trips and both floor regimes/views. Existing portfolio golden results remain regression evidence with no new relief. Synthetic eligibility references are fictional examples, never customer evidence.

Primary sources

- CRR, Articles 501 and 501a, consolidated 2026-01-01
- EBA Q&A 2020_5551: both factors may apply when both sets of conditions hold

Source documents are not bundled. This correction does not automatically explain differences to a particular bank's published RWA. See supporting-factor documentation for inputs, eligibility, compatibility and audit output.